Visualizing Point-to-Plane (PL-ICP) Solving Steps (TikZ Series)

Step 1: Identify Correspondences & Target Normals (n_i)

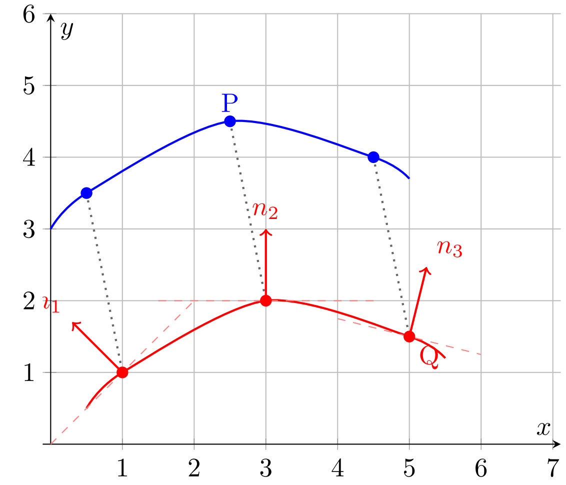

Figure 1: Step 1: Point clouds P (source) and Q (target) are matched. In PL-ICP, we must compute the local surface normal vectors n_i and tangent planes (dashed lines) for every point in the target set Q.

Step 2: Formulate the Error Metric (Zoomed View)

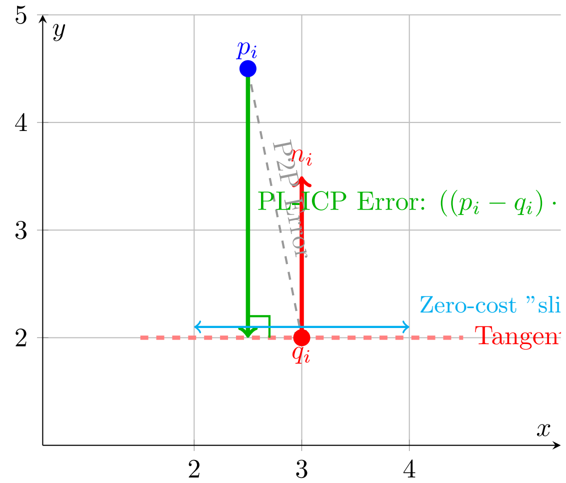

Figure 2: Step 2: The core difference. Standard ICP penalizes the direct distance to q_i (dashed black line). PL-ICP only penalizes the perpendicular distance to the tangent plane (solid green line), allowing points to freely slide horizontally, reducing local minima.

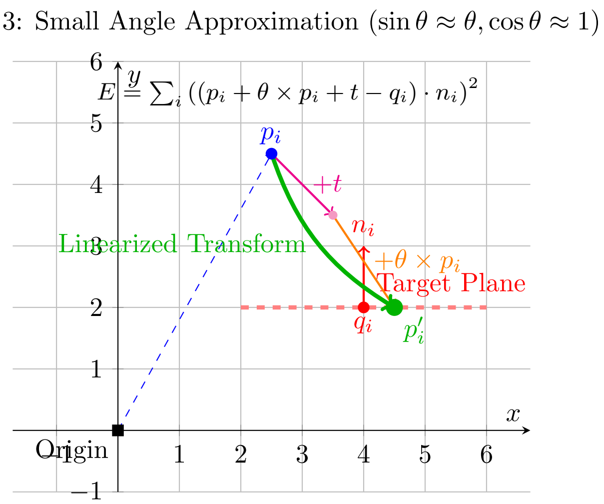

Figure 3: Step 3: Because the rotation matrix is non-linear, we linearize the transformation by assuming the rotation angle θ is small. The movement of p_i is decomposed into a translation vector t and a cross-product $\theta \times p_i$. We constrain this new point p'_i to lie exactly on the target plane.

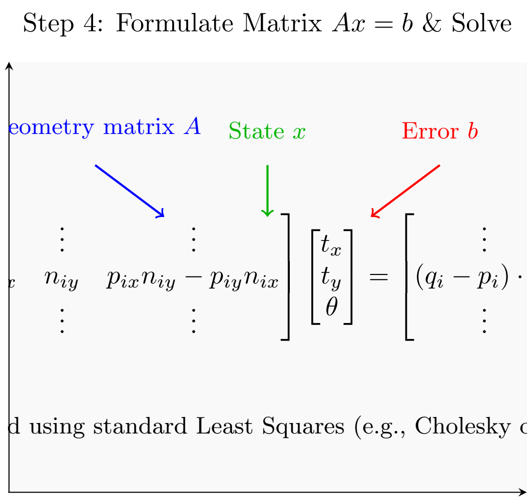

Figure 4: Step 4: Stacking the constraints from all points creates a massive over-determined linear system $Ax = b$. Solving this yields the optimal translation components (t_x, t_y) and the rotation angle (θ) simultaneously, bypassing the centroid calculations required in P2P-ICP.

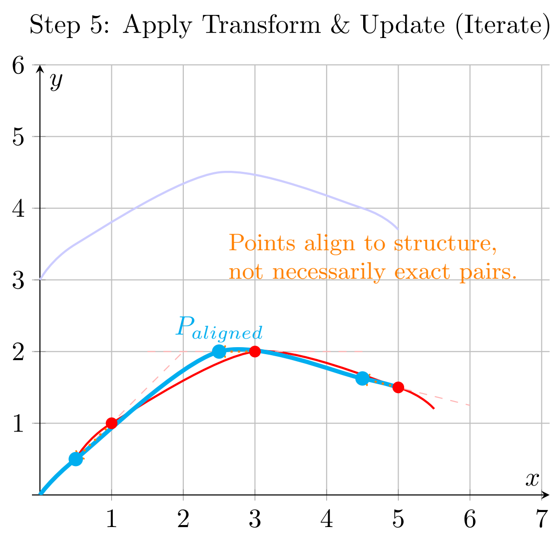

Figure 5: Step 5: Visualizing the final aligned point cloud (cyan). Because the metric minimized point-to-plane distance, the cyan points map exactly onto the mathematical surfaces defined by Q's normals, but they do not need to correspond exactly to the discrete red points (q_i). This is the "sliding" characteristic of PL-ICP.